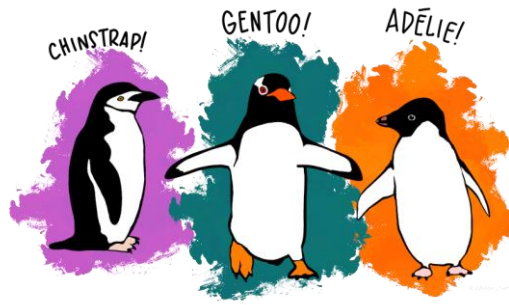


COP2800C Module 1 Graded Programming Assignment

Palmer Penguins

- In this course we will design Java programs which manipulate a dataset which represents the "Palmer Penguins", a research project which tracked data about three penguin species observed on three islands in the Palmer Archipelago, Antarctica: Chinstrap, Gentoo, and Adélie.



<https://allisonhorst.github.io/palmerpenguins/>

- Historically there have been some well -known and well -used (overused) datasets used for Data Science

MNIST (1995)



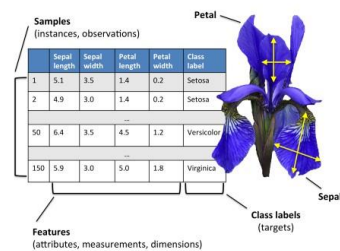
Fashion MNIST (2017)



MTCARS - 32 cars from the 1973 - 74 model year. Statistics about 1974 cars: fuel consumption, design, and performance.

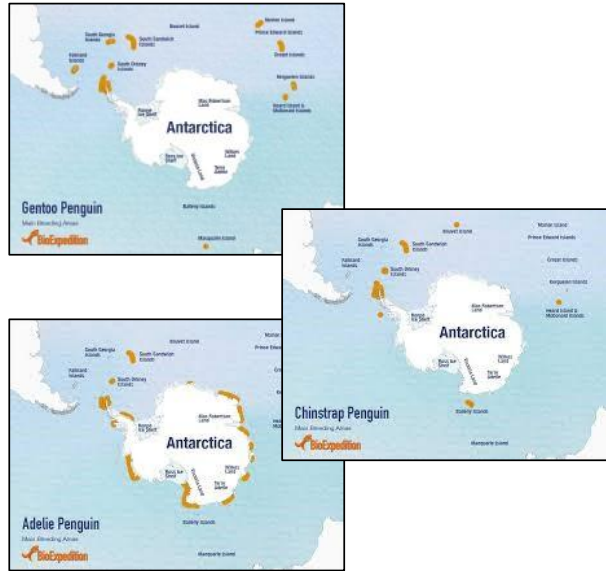


IRIS: Fisher's Iris data set was introduced by the British statistician, eugenicist, and biologist Ronald Fisher in a 1936 paper.



Enter Palmer Penguins!

- Data includes size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica.
- Collected from 2007- 2009 by Dr. Kristen Gorman, Palmer Station Long Term Ecological Research Program, part of the US Long Term Ecological Research Network
- Freely available ("No Rights Reserved").
- "The goal of Palmer Penguins is to provide a great dataset for data exploration & visualization"



Species

a factor denoting penguin species (Adélie, Chinstrap and Gentoo)

Island

a factor denoting island in Palmer Archipelago, Antarctica (Biscoe, Dream or Torgersen)

CulmenLength_mm

a number denoting bill length (millimeters)

CulmenDepth_mm

a number denoting bill depth (millimeters)

FlipperLength_mm

an integer denoting flipper length (millimeters)

BodyMass_g

an integer denoting body mass (grams)

Sex

a factor denoting penguin sex (female, male)

Year

an integer denoting the study year (2007, 2008, or 2009)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	StudyName	SampleNumber	Species	Region	Island	Stage	ID	ClutchCompletion	EggDate	CulmenLength-mm	CulmenDepth-mm	FlipperLength-mm	Bod Mass-g	Sex
2	PAL0708	1	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N1A1	Yes	11/11/2007	39.1	18.7	181	3750	MALE
3	PAL0708	2	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N1A2	Yes	11/11/2007	39.5	17.4	186	3800	FEMALE
4	PAL0708	3	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N2A1	Yes	11/16/2007	40.3	18	195	3250	FEMALE
5	PAL0708	4	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N2A2	Yes	11/16/2007					
6	PAL0708	5	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N3A1	Yes	11/16/2007	36.7	19.3	193	3450	FEMALE
7	PAL0708	6	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N3A2	Yes	11/16/2007	39.3	20.6	190	3650	MALE
8	PAL0708	7	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N4A1	No	11/15/2007	38.9	17.8	181	3625	FEMALE
9	PAL0708	8	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N4A2	No	11/15/2007	39.2	19.6	195	4675	MALE
10	PAL0708	9	Adelie Penguin (Pygoscelis adelliae)	Anvers	Torgersen	Adult, 1 Egg Stage	N5A1	Yes	11/9/2007	34.1	18.1	193	3475	

This assignment will give you a "quick win" with an easy Java program.

Design a public Java class named "PalmerPenguins" which introduces the user to the penguins in the dataset. You'll need a main module in the class which displays the expected output shown below (pay attention to the formatting).

Declare three static String constants in the class (above the main method) which represent the penguin species. Remember that constant names must always be spelled with all upper case. These strings should contain `_only_` the species name, do not include any surrounding text. Here's an example:

```
static final String SP_CHINSTRAP = "Chinstrap";
```

Declare a static int constant which represents the total number of species.

In your main method, use concatenation with string literals in your output statements to match the formatting in the expected output. Here's an example, where `"\t"` is a tab character:

```
System.out.println("\t" + SP_CHINSTRAP + "!");
```

Use concatenation to insert the static int constant value in the last line.

Be sure your submission uses correct indentation and includes a 4-line ID header at the top of the file:

```
// filename (should be PalmerPenguins.java)
// your name
// submission date
// 1-line summary of program's purpose (_not_ "module 1 assignment")
```

Expected Output:

```
Introducing the Palmer Penguins:
    Chinstrap!
    Gentoo!
and last but not least...
    Adelie!
There are a total of 3 penguin species in this dataset.
```